Vishal Chaudhary
B.Tech Information Technology, NSUT

Email: vishal.chowdhary.ug24@nsut.ac.in
GitHub: github.com/vishal-chaudhary23
LinkedIn: linkedin.com/in/vishal-chaudhary-8996a2366
LeetCode: leetcode.com/u/vishal_chaudhary23

I am a second-year Information Technology undergraduate at NSUT. My interest in machine learning began during my first year while exploring different areas of technology. I was particularly fascinated by how machines learn complex patterns from data using neural networks inspired by the human brain. As I studied concepts such as regression, neural networks and deep learning through self-learning and Andrew Ng's courses, I realized machine learning was a powerful way to solve real-world problems. This realization motivated me to deepen my understanding through projects and practical experimentation.

As my understanding of machine learning grew, I worked on projects across computer vision, natural language processing and generative AI. I developed a Food101 image classification system and an American Sign Language recognition model, which strengthened my understanding of deep learning and computer vision. Also, I built a vehicle detection and tracking system using YOLO and SORT, gaining hands-on experience with object detection and real-time vision applications. I also explored modern AI systems, I built Retrieval-Augmented Generation (RAG) based application using LangChain, and graph databases to build applications that combine large language models with external knowledge sources. In the NLP domain, I developed a disaster tweet classification system for identifying emergency-related messages from social media data. Through these projects, I gained practical experience in data preprocessing, model training, evaluation, and designing AI solutions for real-world problems.

Beyond personal projects, I participated in Kaggle competitions such as the Titanic Survival Prediction challenge. I also explored emerging AI technologies through Generative AI and Model Context Protocol (MCP) workshops, broadened my understanding of modern AI systems. Alongside this, I have solved over 250 problems on LeetCode and participated in programming contests, strengthening my analytical thinking and ability to approach complex problems systematically. Maintaining my projects on GitHub has further helped me develop disciplined experimentation and documentation.

While my projects have helped me build a strong practical foundation in machine learning, most of my experience has been limited to academic and personal projects. I am eager to understand how machine learning is applied to solve large-scale business problems in industry. In particular, I want to learn how experienced researchers and scientists approach problem formulation, select appropriate models, evaluate trade-offs and deploy solutions that operate reliably at scale. Learning directly from Amazon Scientists would provide valuable insights into the technologies, frameworks and engineering practices used in real-world machine learning systems ---- an exposure that is difficult to gain through self-learning alone.

Amazon ML Summer School presents an ideal opportunity to bridge this gap. The program's unique combination of advanced machine learning concepts and insights from Amazon scientists particularly appeals to me. I am excited by the opportunity to learn from experts who have worked on large-scale systems and real-world applications across domains such as recommendations, search and generative AI. Beyond strengthening my theoretical foundations, I hope to gain a deeper understanding of how machine learning solutions are designed, evaluated and deployed to address complex business challenges. I believe the knowledge, mentorship and exposure offered by AMLSS will play a significant role in shaping my growth as a machine learning practitioner.

As someone who enjoys learning through exploration, building and continuous improvement, I believe Amazon ML Summer School will provide the technical depth, industry perspective and mentorship needed to take the next step in my AI journey. I look forward to learning from the program, engaging with like-minded peers, and applying these insights to future projects, research and impactful machine learning solutions.